

# Case Study: Cycling Traffic Data in Flanders

*Group 4*

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# 1 Introduction

## 1.1 Background and Motivation

Cycling is a cornerstone of sustainable mobility in Flanders, yet it remains disproportionately dangerous. In Flanders, cyclists accounted for 30% of all road fatalities in 2024 — the second largest group after car occupants — while fatalities among all other road user types have declined significantly since 2005, cyclist deaths have increased by 15% over the same period [Statistiek Vlaanderen, 2024].

To monitor cycling infrastructure, the Flemish road agency Agentschap Wegen en Verkeer (AWV) operates 151 automatic bicycle counting stations across Flanders (*fietstellingen*), recording cyclist counts at 15-minute intervals. AWV also maintains a *Gevaarlijke Punten* (dangerous points) list identifying locations with a history of serious accidents [Agentschap Wegen en Verkeer, 2025]. However, this list is inherently reactive: a location must first accumulate recorded accidents before being flagged, and it does not account for cycling exposure — a busy junction with ten accidents may be safer per cyclist than a quiet path with two.

## 1.2 Problem Statement

This project addresses that gap by developing a predictive cycling risk assessment model for AWV sensor locations. The model uses lagged traffic volume, weather conditions, and accident history from the preceding months to forecast whether a location will experience elevated risk in the following month. Because all features are drawn from the period prior to the prediction window, the model is genuinely forward-looking and avoids data leakage from concurrent observations. The analysis is scoped to 95 grouped station entities derived from 151 individual AWV EcoCounter sensors, which provide the exposure data necessary to compute normalized risk estimates.

## 1.3 Research Objectives

1. Produce reliable next-month crash risk predictions for AWV station groups across Flanders.
2. Generate historical and forward-looking safety rankings, identifying station groups where accident risk is disproportionate to cycling exposure.
3. Develop an interactive web dashboard that visualises historical and forecast safety rankings across AWV station groups, and provides a station-level prediction engine allowing users to adjust traffic and weather inputs to simulate next-month crash risk.

## 2 Data Sources and Preprocessing

### 2.1 Data Sources

#### 2.1.1 Cycling Traffic Data (Fietstellingen)

The *Fietstellingen* dataset comprises monthly count records (`data-YYYY-MM.csv`) containing 15-minute interval cyclist counts per sensor, a site metadata file (`sites.csv`) providing the geographic coordinates, municipality, and installation date of each EcoCounter station, and a directional configuration file (`richtingen.csv`).

#### 2.1.2 Historical Accident Data

The *Geolocation of Road Traffic Accidents 2017–2024* dataset published by Statbel [Statbel, 2025] records all police-reported road traffic accidents in Belgium with geographic coordinates. Each record includes the year and month of collision, road user types, weather and road conditions at the time of the accident, and the municipality of occurrence.

#### 2.1.3 Weather Data (Open-Meteo)

Hourly weather observations for each sensor location were retrieved from the Open-Meteo Historical Archive API [Open-Meteo, 2024] using each sensor’s geographic coordinates and covering the period from its installation date onwards. Variables include temperature, relative humidity, precipitation, snowfall, cloud cover, wind speed, wind direction, and weather code, all reported at hourly resolution in Brussels local time.

### 2.2 Preprocessing Pipeline

Data quality checks identified three sensors requiring special treatment: sensor 144 recorded only one month of data with 100% missing counts, sensor 142 was identified as a test sensor with a 10.9% null rate, and both were excluded from analysis. Sensor 123 exhibited a null rate of 76.8% and is retained but treated with caution. A broader spike in missing values between March and July 2022 across multiple sensors suggests a network-wide data collection issue during this period; affected records are dropped prior to modelling.

#### 2.2.1 Temporal Alignment

All data sources are aligned to the period September 2019 to December 2024. The lower bound corresponds to the earliest available fietstellingen count data in this study, while the upper bound is determined by the coverage of the Statbel accident dataset, which extends to December 2024.

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### 2.2.2 Geographic Filtering and Category Refinement

Accident records are filtered to retain only those involving at least one cyclist as a road user and falling within the geographic bounding box of the AWW sensor network.

### 2.2.3 Spatial Buffer Join

Accident coordinates, originally provided in the Belgian Lambert projection (EPSG:31370), are converted to WGS84 (EPSG:4326) prior to spatial joining. Of the filtered accident records, 10.3% lack valid coordinate information and are excluded from the spatial join.

The remaining records are matched to sensors using a BallTree radius search within a 1km threshold. A threshold of 1km was selected over the stricter 500m radius, which yielded only 908 matched accidents with 109 of 151 sensors having fewer than 10 matched crashes — insufficient signal for monthly modelling. At 1km, 2,734 accidents are matched across 148 sensors, with only 53 sensors having fewer than 10 matched crashes.

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### 2.2.4 Station Grouping

As the *fietstellingen* network frequently deploys multiple sensors at the same physical location — such as separate inbound and outbound counters at the same junction — individual sensors are consolidated into station groups, forming the unit of analysis for all subsequent modelling. Two sensors are merged into the same group if they share the same municipality, have matching normalized names, and are located within 200 metres of each other. Traffic and crash features are aggregated at the group level, with the group's geographic position set to the centroid of its member sensors. Accident records are deduplicated within each group to ensure that a single crash falling within range of multiple member sensors is counted only once. This process reduces the 151 individual sensors to 95 grouped station entities.

## 3 Exploratory Data Analysis

### 3.1 Traffic Patterns

Figure 1 shows the distribution of total cyclist counts by hour of day across all sensors and the full study period. Cycling activity exhibits two clear peaks: a morning peak at 08:00 and an afternoon peak at 16:00–17:00, consistent with commuting behaviour. This bimodal pattern confirms that cycling activity in Flanders is predominantly commuting-driven, with consistent morning and evening peaks across the study period.

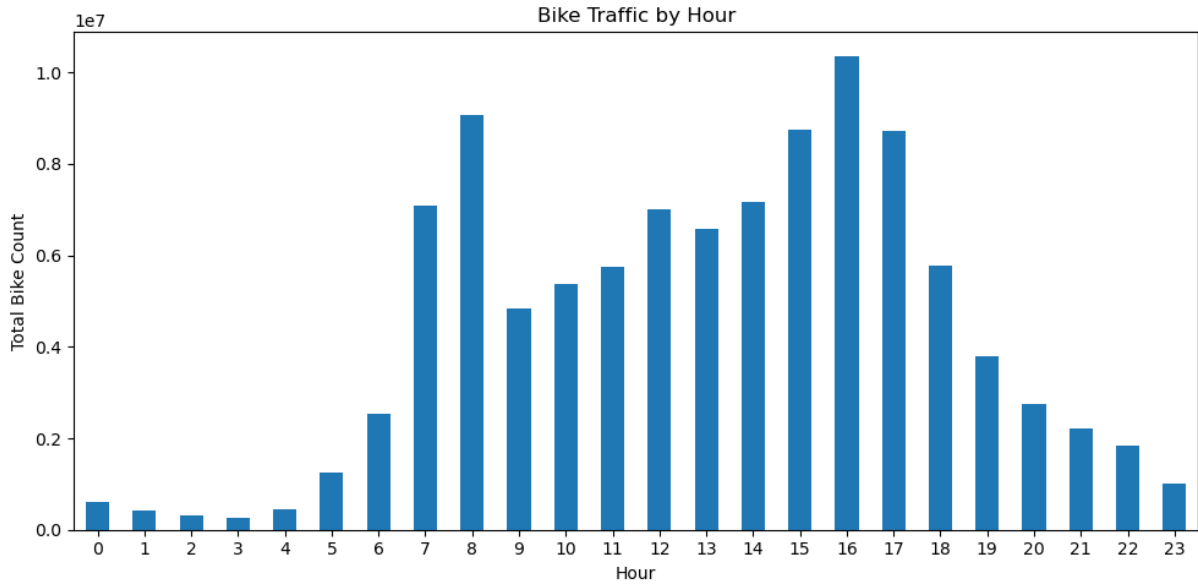


Figure 1: Total cyclist counts by hour of day across all sensors (2019–2024).

Figure 2 shows that weekday traffic is substantially higher than weekend traffic, with Tuesday and Wednesday recording the highest counts. Saturday and Sunday show noticeably lower counts than weekdays, reflecting the dominance of commuting trips in the dataset.

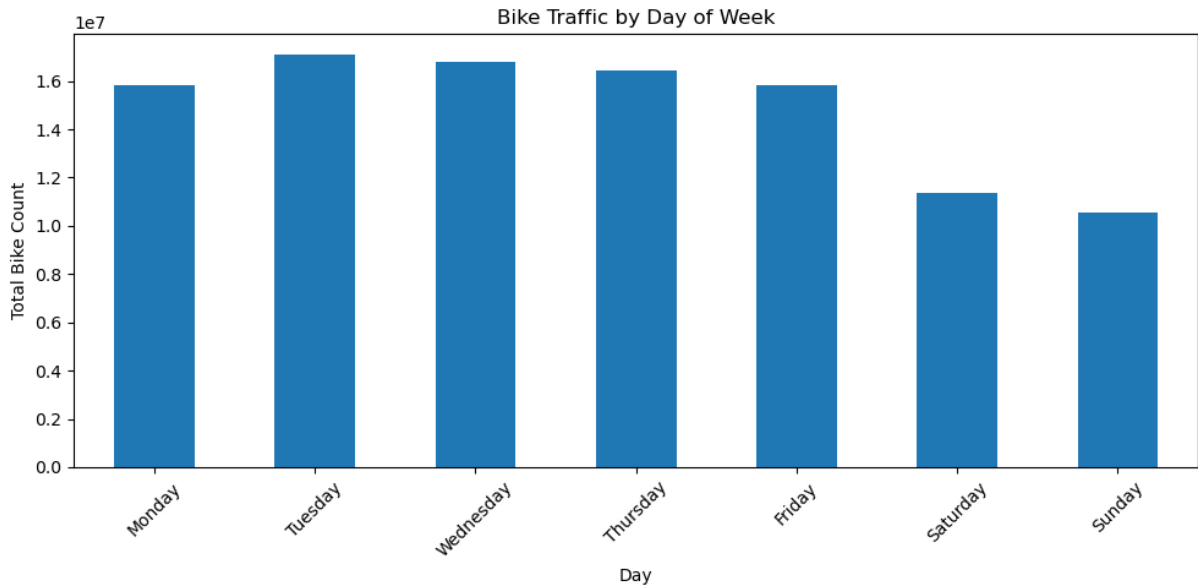


Figure 2: Total cyclist counts by day of week across all sensors (2019–2024).

### 3.2 Accident Characteristics

Figure 3 shows the temporal distribution of bicycle accidents in Flanders across year, month, and hour of day.

Annual accident counts remain broadly stable between 2017 and 2024, ranging from ap-

proximately 6,500 to 8,400 per year, with a notable peak in 2022 and 2023. The absence of a clear downward trend suggests that cycling safety in Flanders has not improved significantly over the study period despite broader road safety efforts.

Monthly distribution reveals a strong seasonal pattern, with accidents peaking in June and September and reaching their lowest levels in January and February. This reflects the higher cycling activity during warmer months and supports the inclusion of seasonality features in the model.

The hourly distribution mirrors the commuting pattern observed in traffic flow, with clear peaks at 08:00 and 16:00–17:00. This alignment between high traffic volume and high accident frequency during peak commuting hours underscores the relevance of traffic volume as a predictor of crash occurrence.

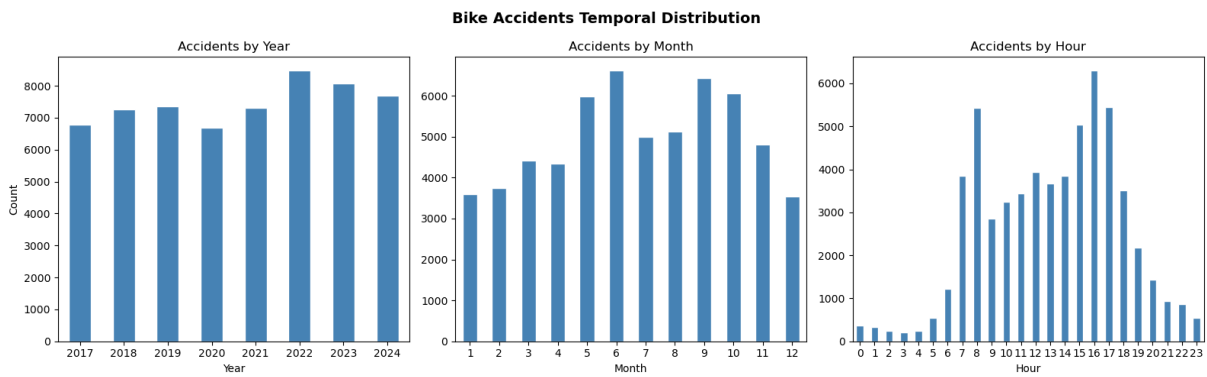


Figure 3: Temporal distribution of bicycle accidents in Flanders by year, month, and hour of day (full dataset 2017–2024; analysis period restricted to 2019–2024).

### 3.3 Environmental Conditions at Time of Accident

Analysis of weather and road surface conditions at the time of accident shows that the majority occurred under favorable circumstances: 83.1% under normal weather and 74% on dry road surfaces. Although these distributions may appear counter-intuitive, they likely reflect higher cycling exposure under favorable conditions rather than lower inherent risk during adverse weather or wet roads. As adverse conditions account for a non-trivial share of accidents, 6.7% during rainfall and 12.9% on wet road surfaces, weather variables are included as features in the predictive model.

Regarding light conditions, 79% of accidents occurred during daylight, consistent with the hourly traffic and accident patterns identified in Sections ?? and ??, where both cycling volume and accident frequency peak during daytime commuting hours.

Accidents are roughly equally distributed between intersection (44.5%) and non-intersection locations (55.5%).



## 4 Methodology

### 4.1 Data Aggregation and Monthly Dataset Construction

The unit of analysis for this study is the station-month: one row per grouped station entity per calendar month. Three data streams are independently aggregated to this level before being merged into a single modelling dataset.

Traffic data are aggregated from 15-minute interval counts to monthly summaries per station and direction (inbound and outbound), yielding features including total cyclist count, mean and maximum interval counts, and the number of zero-count intervals.

Hourly weather observations are aggregated to monthly summaries per station, producing features including mean, minimum, and maximum temperature, total precipitation, total snowfall, average cloud cover, and average wind speed.

Matched accident records are counted per station per month to produce two crash outcome variables: `bike_crash_count_1km_grouped`, the number of bicycle accidents matched within 1km of the station in that month, and `bike_crash_happened_1km_grouped`, a binary indicator of whether at least one such accident occurred.

An inner join between traffic and weather ensures only station-months with complete observations from both sources are retained. The dataset is further restricted to months in which at least one bicycle accident was recorded within the study area; in practice, this condition is satisfied for all months in the 2019–2024 study period. The final forecasting dataset comprises 2,733 station-month rows across 95 grouped station entities.

### 4.2 Target Variable Definition

The primary prediction target is a binary indicator of whether at least one bicycle accident occurs within 1km of a station group in the following calendar month:

$$y_{i,t} = 1 [\text{bike\_crash\_count\_grouped}_{i,t+1} > 0] \quad (1)$$

where  $i$  denotes the station group and  $t$  the current month.

Each row in the forecasting dataset represents the state of a station group at month  $t$ , with features derived from months prior to  $t$ , and the target reflecting outcomes in month  $t + 1$ . This temporal structure ensures all features are available at prediction time.

### 4.3 Feature Engineering

For each of the three feature sources — traffic, weather, and accident history — the core monthly values are transformed using one- and two-month lags, three- and six-month rolling means, and three-month rolling standard deviations.

- One- and two-month lags
- Three- and six-month rolling means
- Three-month rolling standard deviation

All transformations use only values from months prior to the prediction month to prevent data leakage.

**Traffic features** include monthly inbound and outbound cyclist counts, mean interval counts, zero-count intervals, month-on-month changes in flow, inbound-to-outbound ratio, and expanding historical averages per station group.

**Weather features** include monthly temperature, precipitation, wind speed, and cloud cover, along with month-on-month changes in temperature and precipitation.

**Accident history features** include the grouped monthly crash count within 1km, a binary indicator of crash occurrence, crash rate per 1,000 cyclists, three- and six-month rolling crash sums, and an expanding historical average crash count per station group.

**Seasonality** is encoded using sine and cosine transformations of the month of year to capture its cyclical nature.

## 4.4 Model Selection and Training

CatBoost is selected as the primary classifier based on its strong performance on tabular data with mixed feature types and its native handling of categorical variables and missing values. Hyperparameters are tuned using Optuna over 40 trials on the grouped station dataset, optimizing for ROC AUC.

A three-way time-based split is applied: the earliest 60% of months form the training set (September 2019 – September 2022), the following 20% form the validation set (October 2022 – September 2023), and the remaining 20% form the held-out test set (October 2023 – December 2024). Hyperparameter tuning is performed on the validation set, and the final model is retrained on the combined training and validation data using the optimal parameters before evaluation on the held-out test set.

## 4.5 Threshold Tuning

CatBoost outputs a predicted probability of crash occurrence for each station group. A decision threshold is selected by searching over values from 0.05 to 0.95 in steps of 0.01 and choosing the threshold that maximizes F1 score on the validation set. This approach balances precision and recall, which is preferable to the default threshold of 0.5 given the class imbalance in the dataset (30.5% positive target share). The selected threshold of 0.41 is then applied to the held-out test set without further adjustment, ensuring an unbiased final evaluation.

## 4.6 Evaluation Metrics

Model performance is evaluated using the following metrics on the held-out test set:

- **ROC AUC:** measures the model's ability to rank high-risk station-months above low-risk ones, independent of the classification threshold.
- **F1 score:** the harmonic mean of precision and recall, capturing the balance between false positives and false negatives at the selected threshold.
- **Precision:** the proportion of predicted high-risk station-months that are truly high-risk.
- **Recall:** the proportion of truly high-risk station-months that are correctly identified by the model.
- **Confusion matrix:** a breakdown of true positives, false positives, true negatives, and false negatives at the selected threshold.

Accuracy is also reported; however, it is not the primary criterion given the minority positive class.

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## 5 Results

### 5.1 Model Comparison

### 5.2 Threshold Tuning Results

### 5.3 Feature Importance Analysis

### 5.4 Historical Safety Rankings

### 5.5 Forecast Safety Rankings

## 6 Web Application

### 6.1 Interactive Risk Map

### 6.2 Historical and Forecast Safety Rankings

### 6.3 Station-Level Detail Panel

### 6.4 Prediction Engine

## 7 Discussion and Conclusion

### 7.1 Interpretation of Results

### 7.2 Radius Selection

### 7.3 Monthly versus Quarterly Forecasting

### 7.4 Limitations

The majority of recorded accidents (69.6%) occurred within built-up areas, whereas the AWW sensor network covers a broader range of road types. This spatial mismatch contributes to the low accident-to-sensor match rate observed in the spatial join.

### 7.5 Conclusion and Future Work

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- A    **GitHub Repository**
- B    **Environmental Setup and Key Dependencies**
- C    **Full Feature List**
- D    **Additional Figures**